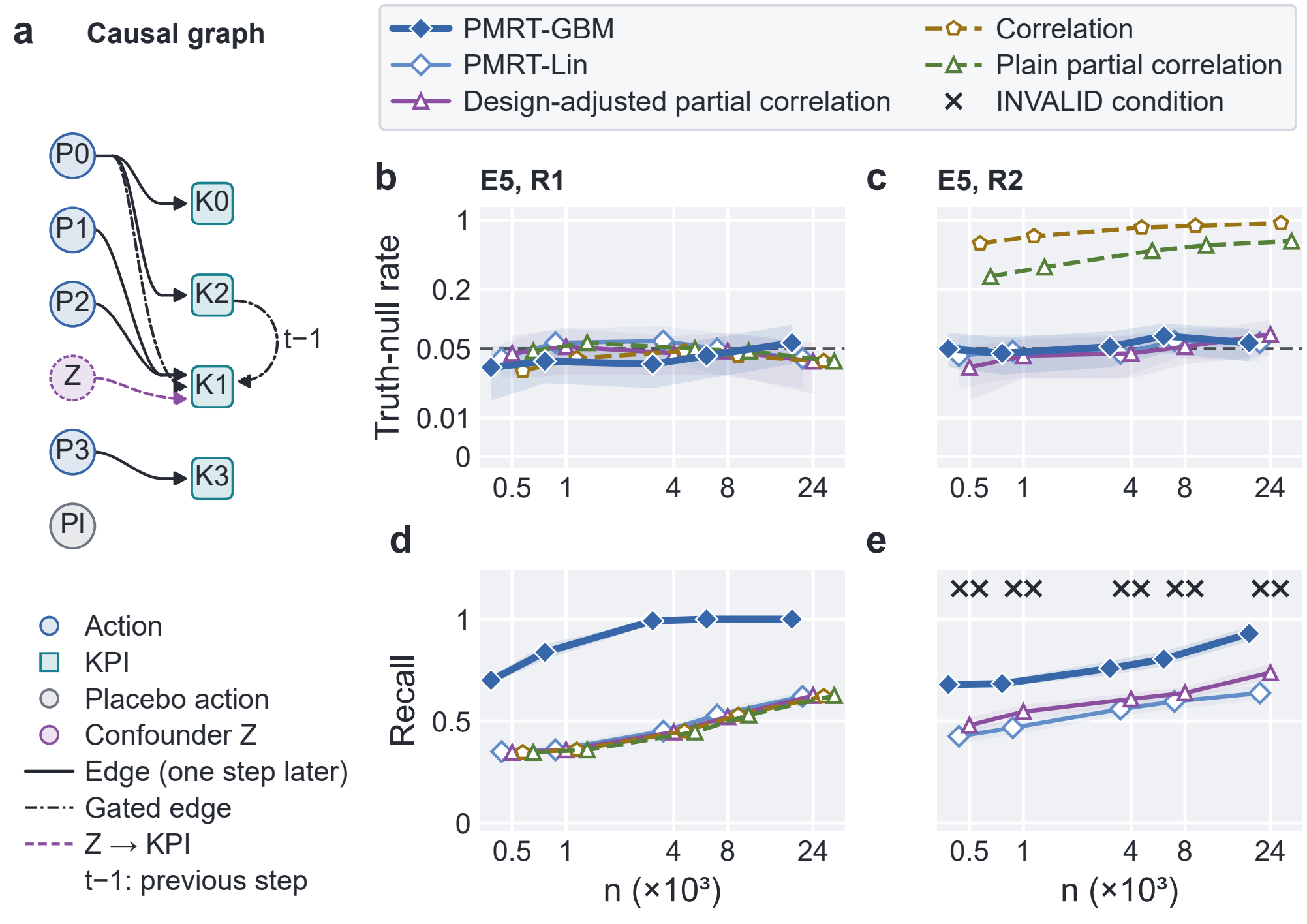

a
Causal graph
P0
P1
P2
Z
P3
PI
K0
K2
K1
K3
t−1
Action
KPI
Placebo action
Confounder Z
Edge (one step later)
Gated edge
Z → KPI
t−1: previous step
PMRT-GBM
PMRT-Lin
Design-adjusted partial correlation
Correlation
Plain partial correlation
INVALID condition
b
E5, R1
c
E5, R2
Truth-null rate
1
0.2
0.05
0.01
0
0.5
1
4
8
24
d
e
Recall
1
0.5
0
n (×10³)